AMAN KUMAR

Data Engineer • AI/ML Specialist • MCP & Agentic AI

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PROFESSIONAL SUMMARY

Results-driven Data Engineer with 3+ years of experience building AI-powered data systems, agentic pipelines, and large-scale ETL infrastructure. Proven track record delivering 95–99% accuracy improvements, ₹30L+ annual cost savings, and multi-million-scale automation. Currently exploring Model Context Protocol (MCP) and agentic AI architectures to build next-generation autonomous data systems.

EXPERIENCE

Data Engineer — Alternative Path *Dec 2025 – Present | Remote*

Stack: *Python, Airflow, Docker, Kubernetes, AWS S3, Lambda, Snowflake, PostgreSQL, MCP*

- Architected end-to-end data pipelines scraping and ingesting data from multiple external sources at production scale.
- Orchestrated automated workflows using Apache Airflow with AWS Lambda triggers and S3-based intermediate storage.
- Designed ingestion pipelines to process, validate, and load structured data into Snowflake and PostgreSQL with high reliability.
- Containerized Airflow scraping workflows on Kubernetes Pods using Docker, enabling isolated and parallel execution at scale.
- Pioneering MCP (Model Context Protocol) integrations to expose internal data tools as AI-accessible services for agentic workflows.
- Partnered with cross-functional teams (product, finance, analytics) to expand and diversify P&L data streams.

Data Analyst — JustDial (Reliance) *Mar 2024 – Nov 2025 | Mumbai*

Stack: *Python, SQL, Azure, GPT/LLMs, OCR, NLP, Spacy, Web Crawling*

- Standardized and cleaned 400K+ records using OCR, web crawling, and fuzzy matching — achieving 98%+ data accuracy.
- Slashed manual validation effort from 20% to 12% by deploying AI/ML and NLP workflows, saving ₹30L annually.
- Engineered a high-throughput web crawler that processed 6M+ websites and extracted 3.2M structured records at 96% accuracy.
- Elevated JustDial category classification precision by 30%+ leveraging GPT-4, LLM prompt engineering, and Spacy NLP pipelines.
- Launched an AI-powered CV parser (OCR + LLM) to automate resume screening and intelligent candidate segmentation.

Data Engineer Intern — GroupM, WPP *Aug 2022 – Feb 2023 | Gurugram*

Stack: *Python, Snowflake, Shell Script, AWS*

- Processed ETL workflows for 5M+ records and tuned SQL pipelines on Snowflake for performance and cost efficiency.
- Accelerated business decision-making efficiency by 15% by analyzing and implementing critical data transformation rules.
- Enriched address datasets using regex patterns, pin code validation, fuzzy matching, and third-party data augmentation.

KEY PROJECTS

AI JD Category Classifier — |

- Attained 99%+ classification accuracy using Spacy NLP combined with LLM-focused prompt engineering techniques.
- Cut misclassification rates by 50% and elevated overall precision by 30%+ across category taxonomies.

Web Crawling & Data Extraction Pipeline — |

- Deployed a distributed crawler to scan 6M+ websites and extract social media URLs, phone numbers, and emails.
- Produced a structured dataset of 3.2M cleaned records at 96%+ accuracy, enabling downstream analytics and enrichment.

End-to-End Airflow ETL Pipeline — |

- Delivered a high-throughput scraping pipeline with Airflow-orchestrated DAGs and AWS S3 storage for reliable multi-source ingestion.
- Spearheaded on-call support to resolve critical pipeline failures, ensuring consistent data reliability in production.

AI-Powered CV Parser — |

- Fused OCR and LLM prompting to create an automated resume screening tool with intelligent candidate segmentation.
- Eliminated manual HR screening bottlenecks by automating structured extraction from unstructured PDF resumes.

TECHNICAL SKILLS

Languages: Python, Advanced SQL, Shell Script
AI / ML: LLMs (GPT-4, Claude), NLP, OCR, ML Models, Prompt Engineering, RAG, Agentic AI
MCP & Agents: Model Context Protocol (MCP), Claude Code, AI Agents, Multi-Agent Systems, Tool Use
Cloud & Databases: Snowflake, AWS (S3, Lambda), Azure, PostgreSQL
Data Engineering: ETL/ELT, Apache Airflow, Docker, Kubernetes, Web Scraping, Data Pipelines, Data Modeling
Scraping & Automation: Scrapy, Playwright, Selenium, BeautifulSoup, API Integration
Tools: GitHub, Advanced Excel, VS Code, Claude Code CLI, Jupyter

EDUCATION

B.Tech — NIT Jalandhar *2018 – 2022*